

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 450

Homework Set 7

Problem 1 (20%) You are estimating the size of a storage tank purchased in 1978 (CEPCI = 219) before computerization of records. You know the purchase price of the tank and that the tank was replaced every 10 years (the size of the tank changed due to changes in plant capacity). Purchasing data are shown in the table below (assume 1988 CEPCI = 300). Estimate the capacity of the storage tank purchased in 1978.

Date	Vessel Capacity (in 1000s of gal)	Purchased Cost
1978	?	\$35,400
1988	105	\$45,300
1998	85	\$45,500

Problem 2 (50%) In this problem we investigate the bare, total module and grass roots costs of the toluene hydrodealkylation plant we discussed at the beginning of the semester (shown in TSBW Figure 1.3 and 1.5, as well as Tables 1.5 and 1.7). Use CAPCOST (use CEPCI = 582) to determine the bare costs, total module costs and grass roots costs of the plant; include a screenshot of CAPCOST's equipment summary page as your "calculations". *Notes:* (1) assume pressures given in Table 1.7 are maximum allowable pressures; (2) leave input at defaults for values for which you don't have data; (3) model the reactor as a jacketed non-agitated pipe in series; (4) MDP heat exchangers should be modeled as multiple pipe; (5) ignore the warning regarding volume of the reactor (it's OK to model as two reactors in series); (6) ignore the warning regarding compressor power (it's OK to use CAPCOST's default).

Problem 3 (30%) This problem regards the drying oil process (Unit 500) described in Appendix B in TSBW (relevant process information appears on the following pages). Assume this process has a stream factor of 0.85.

(a, 10%) Estimate the annual cost of operating labor (C_{OL}). Assume that the salary of an operator is \$70,000/yr.

(b, 5%) What is the annual cost of the hexadecanoic acid (ACO) raw material stream fed to the process as Stream 1? Assume ACO costs \$2.00/kg and has a MW of 256.4 g/mol.

(c, 10%) Examine the information provided for P-501. Assuming a cost of electricity of \$0.07/kWh, what is the annual cost of the utility for P-501?

(d, 5%) Regardless of what values you found earlier in this problem, now assume FCI = \$10 million, cost of operating labor (C_{OL}) = \$1 million/yr, cost of utilities (C_{UT}) = \$30,000/yr, cost of raw materials (C_{RW}) = \$24 million/yr, and cost of waste treatment (C_{WT}) = \$0/yr. Estimate the annual cost of manufacturing (COM_d) for the process.

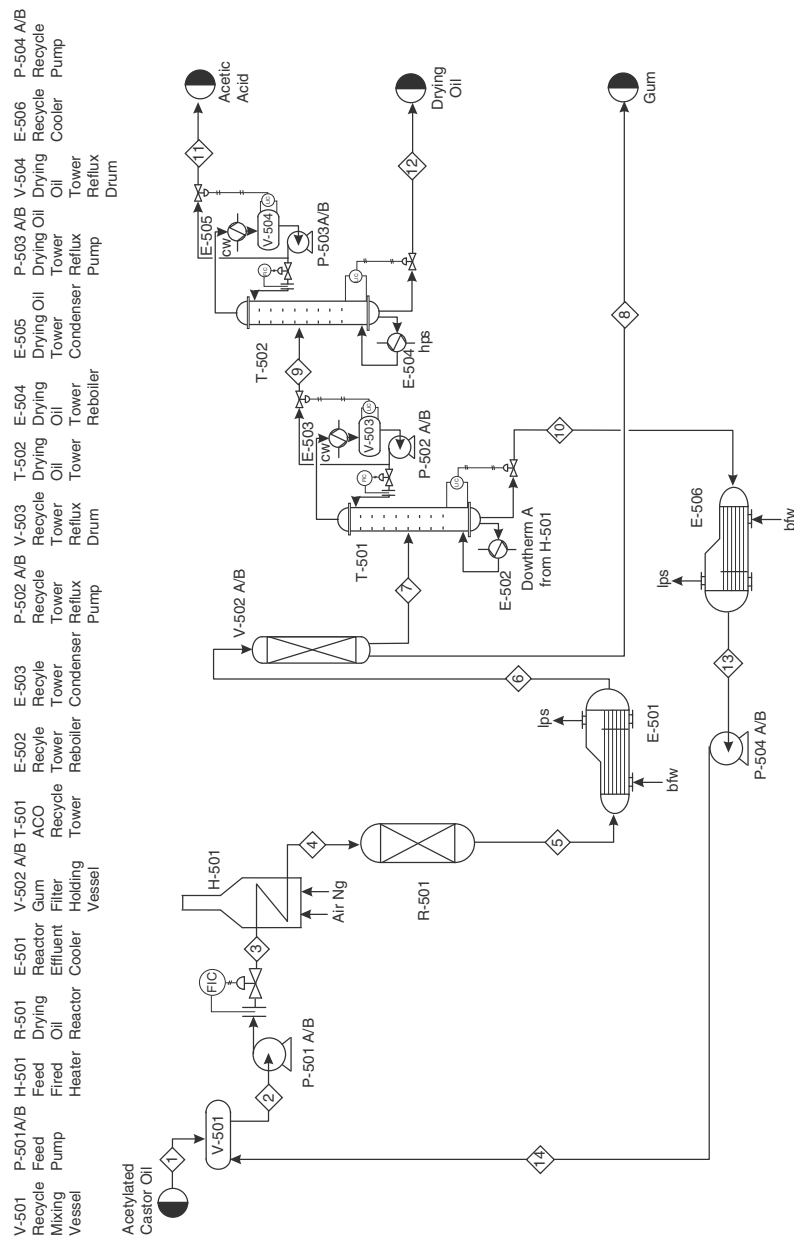


Figure B.4.1 Unit 500: Drying Oil Process Flow Diagram

Table B.4.1 Stream Table for Unit 500

Stream Number	1	2	3	4
Temperature (°C)	25.0	151.0	151.1	380.0
Pressure (kPa)	110.0	105.0	230.0	195.0
Vapor mole fraction	0.00	0.00	0.00	0.00
Flowrate (kg/h)	1628.7	10,703.1	10,703.1	10,703.1
Flowrate (kmol/h)	6.35	41.75	41.75	41.75
Component flowrates (kmol/h)				
Acetic acid	0.00	0.00	0.00	0.00
1-Tetradecene (drying oil)	0.00	0.064	0.064	0.064
Hexadecanoic acid (ACO)	6.35	41.69	41.69	41.69
Gum	0.00	0.00	0.00	0.00
Stream Number	5	6	7	8
Temperature (°C)	342.8	175.0	175.0	175.0
Pressure (kPa)	183.0	148.0	136.0	136.0
Vapor mole fraction	0.39	0.00	0.00	0.00
Flowrate (kg/h)	10,703.1	10,703.1	10,703.1	0.02
Flowrate (kmol/h)	48.07	48.07	48.07	4.61×10^{-5}
Component flowrates (kmol/h)				
Acetic acid	6.32	6.32	6.32	0.00
1-Tetradecene (drying oil)	6.38	6.38	6.38	0.00
Hexadecanoic acid (ACO)	35.38	35.38	35.38	0.00
Gum	4.61×10^{-5}	4.61×10^{-5}	0.00000	4.61×10^{-5}
Stream Number	9	10	11	12
Temperature (°C)	108.0	344.8	119.2	252.8
Pressure (kPa)	125.0	90.0	105.0	125.0
Vapor mole fraction	0.00	0.00	0.00	0.00
Flowrate (kg/h)	1628.7	9074.4	378.6	1250.0
Flowrate (kmol/h)	12.67	35.40	6.29	6.38
Component flowrates (kmol/h)				
Acetic acid	6.32	0.00	6.28	0.03
1-Tetradecene (drying oil)	6.32	0.06	0.01	6.31
Hexadecanoic acid (ACO)	0.04	35.34	0.00	0.04
Gum	0.00	0.00	0.00	0.00

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Stream Number	13	14
Temperature (°C)	170.0	170.0
Pressure (kPa)	65.0	110.0
Vapor mole fraction	0.00	0.00
Flowrate (kg/h)	9074.4	9074.4
Flowrate (kmol/h)	35.40	35.40
Component flowrates (kmol/h)		
Acetic acid	0.00	0.00
1-Tetradecene (drying oil)	0.06	0.06
Hexadecanoic acid (ACO)	35.34	35.34
Gum	0.00	0.00

Table B.4.2 Utility Summary Table for Unit 500

E-501	E-502	E-503	E-504	E-505	E-506
bfw→lps	Dowtherm A	cw	hps	cw	bfw→lps
2664 kg/h	126,540 kg/h	24,624 kg/h	425 kg/h	5508 kg/h	2088 kg/h

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Table B.4.3 Major Equipment Summary for Unit 500

Fired Heater H-501 Total heat duty required = 13,219 MJ/h = 3672 kW Design capacity = 4000 kW Carbon steel tubes 85% thermal efficiency	
Heat Exchangers E-501 A = 26.2 m² 1-2 exchanger, floating head, stainless steel Process stream in tubes Q = 6329 MJ/h E-502 A = 57.5 m² 1-2 exchanger, floating head, stainless steel Process stream in shell Q = 5569 MJ/h E-503 A = 2.95 m² 1-2 exchanger, floating head, stainless steel Process stream in shell Q = 1029 MJ/h	
E-504 A = 64.8 m² 1-2 exchanger, floating head, stainless steel Process stream in shell Q = 719 MJ/h E-505 A = 0.58 m² 1-2 exchanger, floating head, stainless steel Process stream in shell Q = 230 MJ/h E-506 A = 919 m² 1-4 exchanger, floating head, stainless steel Process stream in tubes Q = 4962 MJ/h	
Pumps P-501 A/B Centrifugal/electric drive Carbon steel Power = 0.9 kW (actual) 80% efficient NPSH_R at design flow = 14 ft of liquid P-502 A/B Centrifugal/electric drive Stainless steel Power = 1 kW (actual) 80% efficient	
	P-503 A/B Centrifugal/electric drive Stainless steel Power = 0.8 kW (actual) 80% efficient P-504 A/B Stainless steel/electric drive Power = 0.3 kW (actual) 80% efficient NPSH_R at design flow = 12 ft of liquid
Reactor R-501 Stainless steel vessel V = 1.15 m ³ 5.3 m long, 0.53 m diameter	

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Towers T-501 Stainless steel 56 sieve trays plus reboiler and condenser 25% efficient trays Total condenser Feed on tray = 32 Reflux ratio = 0.15 12-in tray spacing, 2.2-in weirs Column height = 17 m Diameter = 2.1 m below feed and 0.65 m above feed	
T-502 Stainless steel 35 sieve trays plus reboiler and condenser 52% efficient trays Total condenser Feed on tray = 23 Reflux ratio = 0.52 12-in tray spacing, 2.8-in weirs Column height = 11 m Diameter = 0.45 m	
Vessels V-501 Horizontal Carbon steel L/D = 3 V = 2.3 m ³ V-502 Vertical Stainless steel L/D = 5 V = 3 m ³	
V-503 Horizontal Stainless steel L/D = 3 V = 2.3 m ³ V-504 Horizontal Carbon steel L/D = 3 V = 0.3 m ³	

B.4.4 Reference

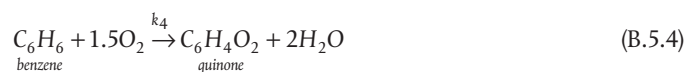
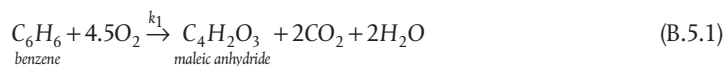
1. Smith, J. M., *Chemical Engineering Kinetics*, 3rd ed. (New York: John Wiley & Sons, 1981), 224–228.

B.5 PRODUCTION OF MALEIC ANHYDRIDE FROM BENZENE, UNIT 600

Currently, the preferred route to maleic anhydride in the United States is via isobutene in fluidized-bed reactors. However, an alternative route via benzene may be carried out using a shell-and-tube reactor, with catalyst in the tubes and a cooling medium being circulated through the shell [1, 2].

B.5.1 Process Description

A process flow diagram for the reactor section of the maleic anhydride process is shown in Figure B.5.1. Benzene is vaporized in E-601, mixed with compressed air, and then heated in a fired heater, H-601, prior to being sent to a packed-bed catalytic reactor, R-601, where the following reactions take place:



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